MILITARY SPECIFICATION

ETHYLENE GLYCOL DINITRATE (EGDN)

1. SCOPE

1.1 This specification covers one type of Ethylene Glycol Dinitrate (EGDN) for use in the formulation of propellants (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-N-3399 - 2-Nitrodiphenylamine (2-NDPA)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-650	- Explosive; Sampling, Inspection and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1376

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(Copies of specifications, standards, drawings and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E-168 - General Technique of Infrared Quantitative Analysis
ASTM-E-300 - Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 0-170
Special Permit No. 2933

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington,D.C. 20402. Orders for the above publications should cite, "49 CFR 0-170 (latest revision)").

3. REQUIREMENTS

3.1 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.2 Material. The EGDN shall contain 0.25% nominal, 2-NDPA, (MIL-D-3399) and shall conform with the properties specified in the following subparagraphs.

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3.2.1 Properties. The ethylene glycol dinitrate (EGDN) shall conform to the properties listed in Table I.

TABLE I

Properties	Percent	Method
Nitrogen, Minimum,min	18.06	4.4.1
Moisture, Maximum, max.	0.25	4.4.2
Acidity as Nitric Acid or Alkalinity as sodium carbonate, max.	0.01	4.4.3
Stability, minutes,min	8.0	4.4.4

3.2.2 Infrared spectrum. The EGDN containing 2-NDPA shall have an infrared spectrum with peaks consistant with those in Figure 1 when tested as specified in 4.4.5. The infrared spectrum shall contain no extraneous significant peaks. A significant peak is defined as a peak having an absorbance greater than 10 percent beyond the normal background.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information

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accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission. Prior to the start of regular production the contractor shall submit a first article sample (see 6.1) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of ethylene glycol dinitrate in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period of excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect production

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uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examination or tests specified in 4.3 and 4.4 of this specification.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation. A lot shall consist of one or more batches, ethylene glycol dinitrate produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of ethylene glycol dinitrate that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the

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procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans, may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect, and an AQL of 0.65 percent for each Minor defect, except where 100 percent inspection is specified.

4.3.2.1 Drums, prior to filling (Polyethylene liner)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Liner cut, torn or punctured	Visual	01001
Minor:	None defined		

4.3.2.2 Drums, prior to closing (Polyethylene liner)

Categories	Defects	Method of Inspection	
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Liner not completely sealed	Visual	02001
Minor:	None defined		

4.3.2.3 Drum, sealed (metal)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
	101. Weight of contents	Scale	03001
	102. Closure of incomplete or damaged to the extent that contents leaking out	Visual	03002
Minor:	AQL 0.65 percent		

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201. Markings misleading or unidentifiable	Visual	03003

4.3.3 Testing

4.3.3.1 Sampling. The EGDN shall be sampled in accordance with paragraph 7 of ASTM-E300 using a rubber dipper. Two independent representative samples of approximately 2 ounces shall be removed from each batch. The EGDN shall be transferred to a rubber bottle with a rubber stopper (see 6.3). The bottles shall be labelled to show the following:

(a) Source of manufacture.
(b) Plant
(c) Contract or purchase order number
(d) Number of pounds in the lot.
(e) Batch number

If either samples fails to comply with any of the requirements specified the batch shall be rejected. The Samples shall be tested for the following properties:

Tests	Defect	Code No.(see 6.2)
Nitrogen	Major	04001
Moisture	Major	05001
Acidity or alkalinity	Major	06001
Stability	Major	07001
Infrared spectrum	Major	08001

After disposal of the EGDN from the rubber sample bottle, the bottle shall be cleaned as follows (see 6.3).

(a) Rinse thoroughly with acetone
(b) Flush generously with hot tap water (3 minutes under a direct flow is considered adequate)
(c) Rinse with distilled water and dry

4.3.4 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.3 and 4.4. The contractor is responsible for assuring proper calibration

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procedures are followed. Government approval of all inpsection equipment is required prior to its use for acceptance purpose.

4.4 Test Methods and Procedures. The tests in 4.4.1 thru 4.4.3 shall be performed using prescribed analytical procedures for duplicate determinations given in standard analytical textbooks.

4.4.1 Nitrogen. The nitrogen shall be determined in accordance with Method 414.1 of Standard MIL-STD-650.

4.4.2 Moisture. The moisture shall be determined in accordance with Method 101.4 of Standard MIL-STD-650.

4.4.3 Acidity or alkalinity. The acidity or alkalinity shall be determined in accordance with Method 102.3 of Standard MIL-STD-650 using benzene as the solvent.

4.4.4 Determination of stability. The stability shall be determined as follows: Filter a portion of the EGDN sample through two thicknesses "S and S." Number 604 filter paper or equivalent. Transfer a 2 ml portion of the filtered material by means of a pipette, to each of three test tubes which are 5.5 inches long, 0.50 inches internal diameter and 0.62 inches outside diameter. (Care should be taken during the transfer not to leave dropslets of the EGDN on the sides of the test tubes.) Stopper each tube by means of a new, tightly fitting cork through which passes a tightly fitted glass rod equipped with a platinun holder for a strip of standard potassium iodide starch indicator paper. Using a forceps hang on the platinum holder a strip of standard potassium iodide starch indicator paper. The standard test papers shall be furnished by the Naval Propellant Plant through the contracting activity and shall normally be approximately 1 inch long and 3/8 inch wide. Moisten a horizontal section in the upper half of the standard test paper with a 50 percent solution (by volume) of pure glycerin in distilled water. This is conveniently accomplished by dipping a small diameter glass rod into the glycerin water solution, and as the rod is withdrawn, contact is made with the side of the container so as to minimize the volume of the solution adhering to the rod. Draw the rod across the paper strip so as to produce a level and distinct line of demarcation on the lower edge of the wet area. Prepare a blank by

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suspending a moistened strip of the standard test paper in a clean dry tube. Adjust the temperature of the heat tube batch to 82.2 ± 1°C. and insert all four test tubes into the heating solution. The depth of immersion of the test tube into the heating solution should be approximately 2 inches. The batch should be placed in such a position that the test tubes are viewed against a white background illuminated by bright diffused daylight. Note the time of insertion of the tubes into the bath. During the test the line of demarcation on the standard test paper should be 3 inches above the level of the EGDN in the test tube. The line of demarcation in the blank tube is regulated at an equivalent height in the tube. Consider the end point of the test to be the first appearance of any discoloration at the line of demarcation between the wet and dry portion of the test paper in the sample tube which is in excess of the discoloration observed at the same position on the test paper in the blank. Note the time for completion of the test to the nearest minute. Consider the minimum time for the three tubes to represent the heat value of the sample. After the test rinse all rods and tubes with acetone to remove EGDN, wash with warm soapy water, rinse thoroughly with tap water, then with distilled water and dry in a steam oven at approximately 80°C.

4.4.5 Infrared spectrum. Determine the infrared spectrum of the sample of material using an Infrared Spectrophotometer. Preparation of sample and method used shall be in accordance with the instructions in ASTM-E-168 using a thin film of the sample between potassium bromide (KBr) windows and comparing the analysis with the infrared standards shown in figure 1.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C. The EGDN is classified as a high explosive, liquid. The EGDN is shipped as a mixture of 59.75% nominal EGDN, 0.25% nominal 2-nitrodiphenylamine and 40% nominal acetone. The EGDN shall be shipped in accordance with Code of Federal Regulations, 49 CFR 170.15. Special Permit No. 2933 of the Department of Transportation (DOT) Hazardous Materials Regulations.

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5.2 Marking. The marking of interior packages and exterior shipping containers shall be in accordance with instructions contained in Special Permit No. 2933, Interstate Commerce Commission Regulations and in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Provisions for submission of first article samples
c. Acceptance and description sheets for each batch in accordance with MIL-STD-1171.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Toxicity. The EGDN may cause dilation of blood vessels, headache, nausea, cyanosis, and reduce blood pressure. Extreme exposures resulting from inhalation, skin contact or ingestion may result in central nervous system depression, coma, and respiratory paralysis. Drinking of any quanity of the liquids must be avoided. Severe headache shortly after exposure may diminish after development of tolerance. Ventilation should be used to reduce the inhalation hazard. Protective gloves and aprons are helpful in avoiding skin contact. Frequent washing in soap and water will minimize skin absorption. Alcohol or other solvents should not be used on the skin. The threahold limit value ceiling is 0.2 pm.

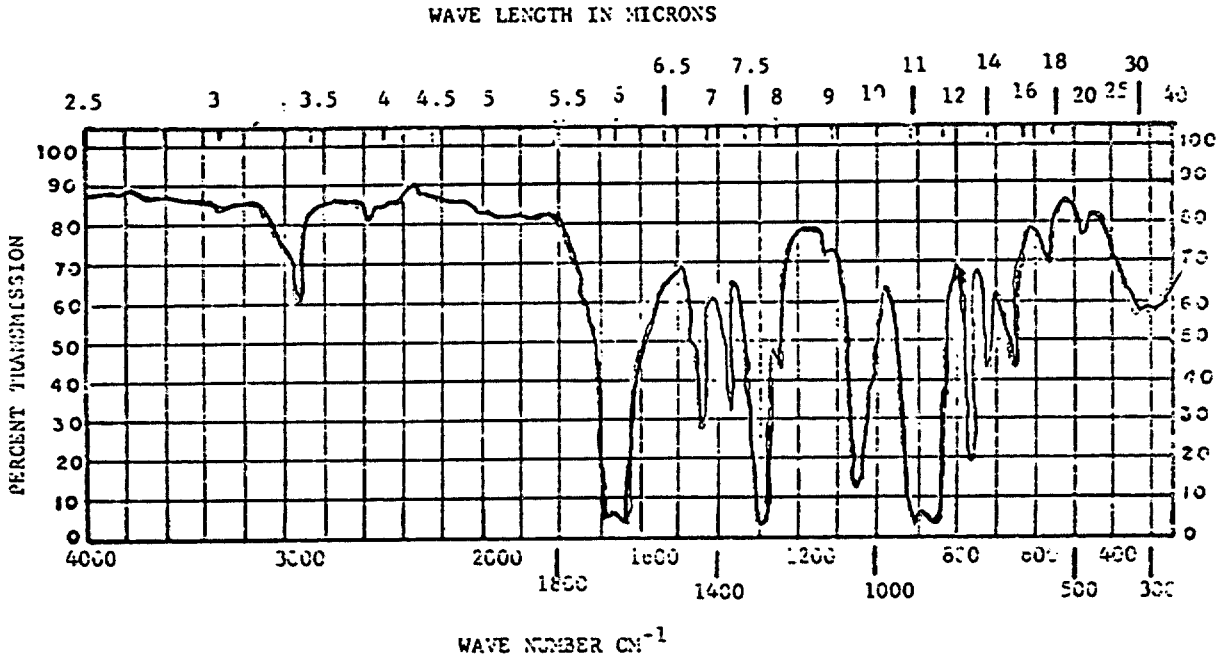
6.4 Intended use. The EGDN of this specification is intended for use as a propellant ingredient.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1376-A009

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ETHYLENE GLYCOL DINITRATE
WITH 0.25% 2-NDPA

FIGURE I

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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